

Logistic Regression

Binary and Multiclass Classification

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Binary Classification

The output belongs to only two categories.

$$\text{Output} \in \{0, 1\}$$

Examples

- Pass / Fail
- Yes / No
- Spam / Not Spam
- Disease / No Disease

Example Dataset

Study Hours	Result
2	Fail
3	Fail
4	Fail
5	Pass
6	Pass
7	Pass
9	Pass

Encode the output as

Fail = 0 Pass = 1

Goal

Study Hours → **Model** → Pass/Fail

Can Linear Regression Solve This Problem?

Suppose we apply Linear Regression. The regression model is

$$\hat{y} = wx + b$$

where

- x : Study Hours
- $\hat{y}$: Predicted Output

The question is

Can Linear Regression correctly classify

Pass and Fail?

Linear Regression Model

Step 2: Fit a linear regression model

The model is

$$\hat{y} = wx + b$$

where w is the slope and b is the intercept.

Using the least-squares formulas,

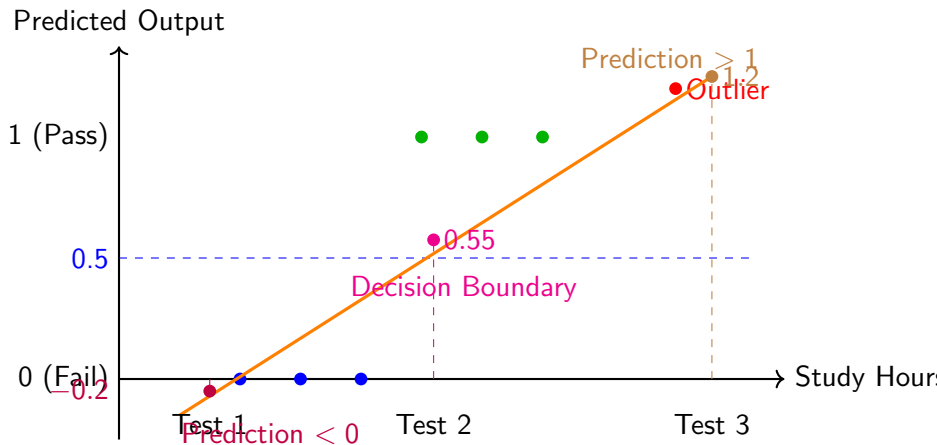
$$w = \frac{m \sum x_i y_i - \sum x_i \sum y_i}{m \sum x_i^2 - (\sum x_i)^2}$$

and

$$b = \frac{\sum y_i - w \sum x_i}{m}.$$

Hence, the fitted linear regression model is $\hat{y} = 0.184x - 0.375$

Why Linear Regression is Not Suitable for Classification



Why Linear Regression is Not Suitable for Classification

Observations

- **Test Point 1:** Linear Regression predicts -0.2 , which is not a valid probability.
- **Test Point 2:** Prediction is around 0.55 , so the class is determined using the threshold 0.5 .
- **Test Point 3:** Linear Regression predicts 1.2 , which is also an invalid probability.
- **Outlier:** The outlier pulls the regression line upward, changing the decision boundary.

Linear Regression is not suitable for Classification.

Logistic Regression

Observation Linear Regression predicts values

$$(-\infty, \infty)$$

which are not valid probabilities. Therefore, we need a function whose output is always between

$$0 \leq P(y = 1|x) \leq 1$$

Logistic Regression achieves this using the **Sigmoid Function**.

$$P(y = 1|x) = \frac{1}{1 + e^{-z}}$$

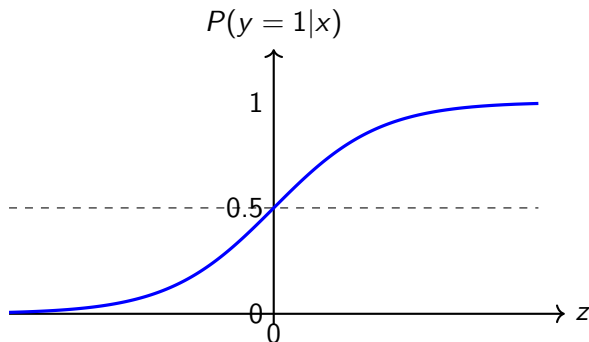
where

$$\begin{aligned} z &= \mathbf{w}^T \mathbf{x} + b \\ &= w_1 x_1 + w_2 x_2 + \cdots + w_n x_n + b. \end{aligned}$$

Thus,

Linear Model $\longrightarrow$ Sigmoid Function $\longrightarrow$ Probability

Sigmoid Function



Properties of Sigmoid Function

- Output is always between 0 and 1.
- Small values of z produce probabilities close to 0.
- Large values of z produce probabilities close to 1.
- At $z = 0$, the probability equals 0.5.

Classification using Logistic Regression

After computing the probability,

$$P(y = 1|x) = \frac{1}{1 + e^{-z}}$$

the model compares it with a threshold.

$$P(y = 1|x) \geq 0.5 \implies \mathbf{Pass (1)}$$

$$P(y = 1|x) < 0.5 \implies \mathbf{Fail (0)}$$

Hence,

Probability $\rightarrow$ Decision Boundary $\rightarrow$ Classification

Why is it called Logistic Regression?

If it performs Classification, why is it called Regression?



Finally,

Probability $\Rightarrow$ Pass / Fail

The regression equation computes

$$z = w^T x + b,$$

while the sigmoid function converts it into a probability. Hence the name **Logistic Regression**.

Comparison: Linear vs Logistic Regression

Linear Regression	Logistic Regression
Continuous Output Predicts any value Output: $(-\infty, \infty)$ Affected by outliers Not suitable for Classification	Categorical Output Predicts probability Output: $[0, 1]$ Less affected Suitable for Binary Classification

Logistic Regression is a Classification Algorithm.

Cost Function Analysis

Recall the Logistic Regression hypothesis

$$h_{\theta}(x) = \sigma(z) = \frac{1}{1 + e^{-z}}$$

where

$$z = \theta_0 + \theta_1 x.$$

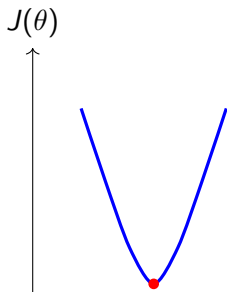
Cost Function Analysis

Linear Regression Hypothesis

$$h_{\theta}(x) = \theta_0 + \theta_1 x$$

Cost Function

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m \left(h_{\theta}(x^{(i)}) - y^{(i)} \right)^2$$

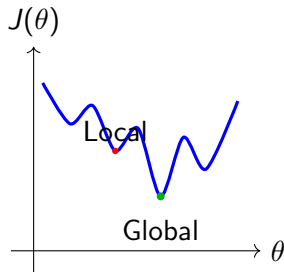


Logistic Regression Hypothesis

$$h_{\theta}(x) = \frac{1}{1+e^{-z}} \text{ where } z = \theta_0 + \theta_1 x$$

Suppose we use

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m \left(h_{\theta}(x^{(i)}) - y^{(i)} \right)^2$$



Why Mean Squared Error is Not Used?

Linear Regression

- The hypothesis is linear.
- The MSE cost function is convex.
- There is only one global minimum.
- Gradient Descent always converges to the optimal solution.

Logistic Regression

- The hypothesis contains the Sigmoid function.
- Using MSE produces a non-convex cost function.
- Multiple local minima may exist.
- Gradient Descent may stop at a local minimum.

Therefore, Mean Squared Error is not used in Logistic Regression.

Cost Function Estimation in Logistic Regression

Step 1: Logistic Regression Hypothesis

$$h_{\theta}(x) = \frac{1}{1 + e^{-z}}$$

where

$$z = \theta_0 + \theta_1 x.$$

Suppose we use the Mean Squared Error (MSE) cost function,

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m \left(h_{\theta}(x^{(i)}) - y^{(i)} \right)^2$$

Since the hypothesis contains the sigmoid function, the MSE cost function becomes non-convex. Therefore, Logistic Regression uses **Log Loss (Cross-Entropy Loss)**.

Cost Function of Logistic Regression

- The logarithm function converts probabilities into a loss value.
- It gives a small loss when the predicted probability is close to the correct class and a very large loss when the model is confidently wrong.
- This property makes the logarithm ideal for defining the cost function in logistic regression.

Let us denote the cost for one training example as

$$\text{Cost} \left(h_{\theta}(x^{(i)}), y^{(i)} \right)$$

$$\text{Cost} \left(h_{\theta}(x^{(i)}), y^{(i)} \right) = \begin{cases} -\log \left(h_{\theta}(x^{(i)}) \right), & \text{if } y = 1 \\ -\log \left(1 - h_{\theta}(x^{(i)}) \right), & \text{if } y = 0 \end{cases}$$

Unified Cost Function

The two cases can be combined into a single equation.

$$\text{Cost} = -y^{(i)} \log \left(h_{\theta}(x^{(i)}) \right) - \left(1 - y^{(i)} \right) \log \left(1 - h_{\theta}(x^{(i)}) \right)$$

Verification

If $y^{(i)} = 1$,

$$\text{Cost} = -\log \left(h_{\theta}(x^{(i)}) \right).$$

If $y^{(i)} = 0$,

$$\text{Cost} = -\log \left(1 - h_{\theta}(x^{(i)}) \right).$$

Hence, a single expression correctly handles both classes.

Cost Function in Logistic Regression

Cost Function

- Logistic Regression uses the **Cross-Entropy Loss** (also called **Log Loss**) to measure the difference between the predicted probabilities and the actual class labels.

$$J(\theta) = -\frac{1}{m} \sum_{i=1}^m \left[y^{(i)} \log(h_{\theta}(x^{(i)})) + (1 - y^{(i)}) \log(1 - h_{\theta}(x^{(i)})) \right]$$

- The objective is to find the parameter vector θ that minimizes the cost function.
- The parameters are optimized using iterative optimization algorithms such as **Gradient Descent**.

Objective of Gradient Descent

In simple linear regression, the hypothesis is

$$h_{\theta}(x) = \theta_0 + \theta_1 x$$

where θ_0 is the intercept and θ_1 is the slope.

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m \left(h_{\theta}(x^{(i)}) - y^{(i)} \right)^2$$

The aim is to minimize the cost function:

$$\min_{\theta_0, \theta_1} J(\theta_0, \theta_1)$$

Gradient Descent Update Rule

Gradient descent updates the parameters as

$$\theta_j := \theta_j - \alpha \frac{\partial J(\theta_0, \theta_1)}{\partial \theta_j}, \quad j = 0, 1$$

Here,

α = learning rate

$\frac{\partial J}{\partial \theta_j}$ = direction of increasing cost

So, we subtract the derivative to move toward the minimum cost.

Update Equations

For θ_0 :

$$\theta_0 := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^m \left(h_{\theta}(x^{(i)}) - y^{(i)} \right)$$

For θ_1 :

$$\theta_1 := \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^m \left(h_{\theta}(x^{(i)}) - y^{(i)} \right) x^{(i)}$$

Repeat these updates until convergence.

Logistic Regression – Example

Question

- The dataset of pass or fail in an exam of 5 students is given in the table.
- Use logistic regression as a classifier to answer the following questions.
- ① Calculate the probability of passing for the student who studied **33 hours**.

Assume the logistic regression model is

Hours Study	Pass (1) / Fail (0)
29	0
15	0
33	1
28	1
39	1

$$z = -64 + 2 \times \text{Hours_Study}$$

Logistic Regression – Solution

1. Calculate the probability of pass for the student who studied 33 hours.

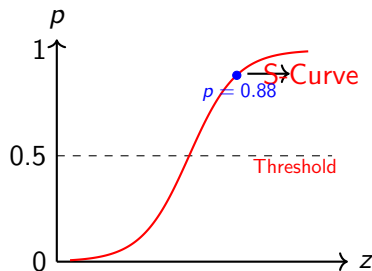
$$p = \frac{1}{1 + e^{-z}}$$

To calculate the probability of passing for the student who studied **33 hours**.

$$z = -64 + 2 \times 33 = 2$$

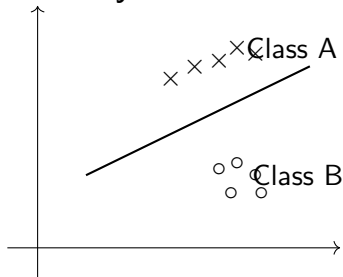
$$\begin{aligned} p &= \frac{1}{1 + e^{-2}} \\ &= \frac{1}{1 + 0.1353} \end{aligned}$$

$$= 0.8808 \approx 0.88$$



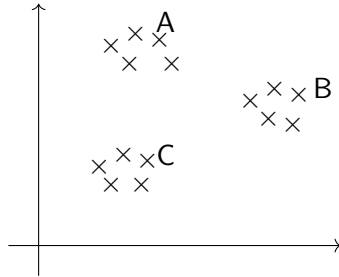
Binary vs Multiclass Classification

Binary Classification



$$y \in \{0, 1\}$$

Multiclass Classification



$$y \in \{A, B, C\}$$

One-vs-Rest Logistic Regression

For multiclass classification, logistic regression can be extended using **One-vs-Rest**.

A, B, C

are converted into three binary problems:

$M_A : A \text{ vs Not } A$

$M_B : B \text{ vs Not } B$

$M_C : C \text{ vs Not } C$

Prediction in Multiclass Logistic Regression

For a new sample x , calculate the probability for each class:

$$P_A(x), \quad P_B(x), \quad P_C(x)$$

Then select the class with the highest probability:

$$\hat{y} = \arg \max_{k \in \{A, B, C\}} P_k(x)$$

Example:

$$P_A(x) = 0.25, \quad P_B(x) = 0.78, \quad P_C(x) = 0.40$$

Since $P_B(x)$ is maximum,

$$\hat{y} = B$$

Multiclass problem is solved using multiple binary logistic models.

Example: Multiclass Prediction

Suppose the three logistic regression models produce the following linear scores:

$$z_A = 0.4, \quad z_B = 1.8, \quad z_C = -0.7$$

Applying the sigmoid function,

$$P_A(x) = \sigma(0.4) = \frac{1}{1 + e^{-0.4}} \approx 0.60$$

$$P_B(x) = \sigma(1.8) = \frac{1}{1 + e^{-1.8}} \approx 0.86$$

$$P_C(x) = \sigma(-0.7) = \frac{1}{1 + e^{0.7}} \approx 0.33$$

Example: Multiclass Prediction

Therefore, the model outputs are

$$P_A(x) = 0.60, \quad P_B(x) = 0.86, \quad P_C(x) = 0.33$$

The predicted class is selected using

$$\hat{y} = \arg \max_{k \in \{A, B, C\}} P_k(x)$$

Since $P_B(x) = 0.86$ is the largest value,

$$\hat{y} = B$$